DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (A)
Osmania University, Hyderabad-07

CLASS TEST-I

Subject : Computer Networks

Class: B.E (CSE and AIML) **Max Marks:** 30
Semester: VI **Duration:** 1.5 hr (12:00 p.m. to 1:30 p.m.)
Academic Year: 2025-26 **Date :** 23-03-2026

	PART A Answer All Questions (5 Q x 2 M= 10 M)	Marks	BT	CO
1. a)	Encode the following bit string using i)**B8ZS** and (ii)**Differential Manchester** Encoding 100000000100001	2	3	1
b)	Encode **QAM** in terms of (i) **4 PSK** and (ii) **8 PSK** for the following data 11 01 11 00 11 10 10 11 00	2	3	1
c)	Explain the **socket address structure** used in computer networks, clearly describing its components, and discuss its **role and usage** in identifying endpoints and enabling communication between client and server processes.	2	2	5
d)	What is a **DNS resource record**? List its common types and explain their typical uses in domain name resolution.	2	2	5
e)	A data packet of **8000 bits** is transmitted through a **5-layer network architecture** in which each layer adds **40 bits of header** and **20 bits of trailer** as overhead. Considering this layered communication process, determine the **total overhead added** to the original message, compute the **total size of the transmitted data**, and calculate the **percentage of bandwidth wasted** due to the added overhead.	2	3	1
	PART B **Answer Any Two Questions (2 Q x 10 M = 20 M)**			
2.	a) Differentiate **circuit switching** and **packet switching** in detail by drawing timing diagrams b) Compare the total delay in transmitting a **2,00,000-bit message** over a **4-hop path** in both a **circuit-switched network** and a **lightly loaded packet-switched network**. The circuit setup time is **3,000 ms**, the propagation delay is **50 ms per hop**, the packet size is **4000 bits**, and the data rate is **40 kbps**. Calculate and compare the **end-to-end delay** for both switching techniques.	5 + 5	2	1
3.	a) Explain the concept of **I/O multiplexing** in socket programming. Describe the different working mechanisms and its advantages. b) Explain how a **concurrent server** is implemented in socket programming, and write a **basic code fragment** to illustrate how multiple clients are handled.	5 + 5	3	5
4	a)A binary data stream **11010110 01101100 11100011 10101010** is encoded using **Base64 encoding**, as used in **Multipurpose Internet Mail Extensions (MIME)** for transmitting binary data over email systems. Determine the **final Base64 encoded output** and write its **equivalent ASCII representation.** b) Explain the **Pulse Code Modulation (PCM)** technique. Describe its working process with a neat diagram. Also, solve a **numerical example** to illustrate how an analog signal is converted into digital form using PCM.	5 + 5	3	5